Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

1. Batch Folder Completeness				
1.1	Batch folder exists with correct naming	Pass	Drive folder 'W3' (1PYKFLRdkZtiQ8sWRgHt8GPb4av7tSNaK, owner garrett.lutz@) holds 'INJ-09-2026-W3 Ops Notes', Scale Tickets/, Testing/, Completed QA Checklists/ (no files yet) and Kerry Oil/ (a separate Kerry removal). Ops 'AECN Rail'!B2 reads INJ-09-2026-W3; Certify period 2026-09-12 to 2026-09-18. [log:drive_folder.root_items, drive_folder.completed_qa_checklists, ops_notes.batch_id_cell, certify.removal_period]	VR App. 1
1.2	Scale Tickets subfolder present and populated	Pass	Scale Tickets/ has one folder per railcar (GPRX5102, SHPX215046, TILX200231) and one per offload (4 + 2 + 4); each offload folder holds a Full and an Empty CAT ticket PDF: 20 files, all opened and read. Certify Sources carries the same 20 files, byte-identical (sha256). [log:drive_folder.scale_tickets_tree, drive_folder.scale_ticket_files, certify.ticket_sources_compare]	VR App. 1
1.3	Testing subfolder present with required lab results	Pass	Testing/ holds 'SR26-05599 W3 QOWV AECN COA.pdf' (page 2 is this removal's AECN certificate SR26-05599.002) and the Kerry certificate SR26-05599.003 (separate removal). [log:drive_folder.testing, lab_results.certificate]	VR App. 1
1.4	Ops Notes spreadsheet present and complete	Flag	Google Sheet 'INJ-09-2026-W3 Ops Notes' (modified 2026-09-29T16:29Z) exported and read with cell comments (none exist): Summary and AECN Rail (3 railcars, 9 BOLs, 10 offload rows, rollup row 46) populated; the AECN Truck tab holds no deliveries. Three gaps: pH blank on every row (5.5); railcar notes M10 and M12 say the mileage was 'intentionally left BLANK' while J/K/L hold 107.5 / 2199 / 42.9 mi (3.5); SHPX215046 H11 'Previously Claimed' is 0 although W2 booked its offloads 3-4 (8.3). [log:ops_notes._primary_source_opened, ops_notes.cell_comments_check, ops_notes.ph_column_G28_G37, ops_notes.railcars, ops_notes.aecn_truck_tab]	VR App. 1
1.5	Injection site confirmed as Basco 6, no Vaulted references	Flag	All 9 BOLs ship to CHARM Industrial Inc./WATCO, SOPOR Baie-Comeau, and every CAT ticket prints 'DELIVERY SITE: BASCO'; the Opelousas-to-Basco map ends at 4687 LA-115, St Landry LA. 'Vaulted' appears 0 times in the 1,338 lines searched (every Ops Notes cell on all 10 tabs, the 10 Certify component names, the 9 BOL ship-to and ship-from fields). One legacy site reference remains: page 5 of each BOL bundle is Charm Purchase Order Q42025 dated 10/2/2025 (line 'Q32025', 6,000 MT RFO) with Ship To 'Charm Industrial, 838 SW Purity Springs Road, El Dorado, KS 67042', Charm's Kansas site, not SOPOR or Basco. [log:vaulted_search, bols.per_bol, bols.po_page, route_evidence.opel_basco_map, scale_tickets.axle_notes]	verify BOLs/Ops Notes/Certify
2. Scale Tickets & Mass Determination				
2.1	Empty and full scale ticket images present for each rail offload	Pass	Each of the 10 offloads has a Full and an Empty CAT ticket in Drive and in Certify Sources (20 of 20 byte-identical); all 20 read from the page image: ticket number, date, time, scale, axle weights, gross, tractor, trailer, commodity. All 20 are linked to 'Mass of Injectant (per injection)' and 'Bio-Oil Mass - AECN via Rail'. [log:scale_tickets.pair_comparison, certify.ticket_sources_compare, certify.attachment_map]	VR Sec 5.1
2.2	Ticket dates within 2 weeks of injection	Pass	Ticket dates run 2026-09-06 10:44 (SHPX215046 offload 1 full) to 2026-09-13 08:36 (TILX200231 offload 3 empty); COBB and Ops date every offload's injection 2026-09-18, 12 days after the earliest weigh. Each empty weigh follows its full weigh. The SHPX215046 offload 1-2 weighs (2026-09-06/07) precede W2's offload 3-4 weighs of the same railcar (2026-09-07 10:52 and 10:55). [log:scale_tickets.pair_comparison, cobb.injection_completion_date, mass_chain.shpx215046_balance.chronological_last_offload]	VR Sec 3.5
2.3	Tractor and trailer match between empty and full tickets, each offload	Fail	9 of 10 pairs match on both identifiers. TILX200231 offload 2 does not: full ticket 1124426254565 (2026-09-11 12:40, scale 1244) tractor CH1, trailer KL720; empty ticket 1312226255533 (2026-09-12 09:14, scale 3122) tractor CH2, trailer KL720. The Ops Notes uses the mismatched empty (29,640 lb) with no cell note, and the September guidance holds no substitution method. Charm's published method for this trigger (August guidance: substitute the trailer standard when '(b) the empty ticket's tractor and/or trailer do not match the Full ticket') gives KL720 29,645 lb and a net of 19,502.18804 kg, 2.26796 kg below the 19,504.456 kg credited. Either apply the KL720 standard with a cell note and state the method in the September guidance, or supply the CH1+KL720 empty weigh for that trip. [log:scale_tickets.pair_comparison.TILX200231_offload_2, vvb_guidance.august_standard_tare, vvb_guidance.no_other_sections, ops_notes.cell_comments_check, ops_notes.offload_table]	VR Sec 5.1

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

2.4	Net mass correctly calculated in Ops Notes (per offload)	Pass	Each offload's net recomputed from its two ticket images as (full lb - empty lb) x 0.453592 equals Ops F28:F37 to five decimals on all 10 rows (mass-chain script scale_ticket_subtraction 10 of 10 PASS; recompute section 1), e.g. GPRX5102 offload 1: (73,140 - 28,760) lb gives 20,130.41296 kg. Ops D/E equal the 20 ticket gross weights. The 10 nets sum to Ops A46 197,321.5918 kg. [log:mass_chain.counts, recompute.lines, ops_notes.offload_table, ops_notes.rollup.sum_cargo_kg, scale_tickets.pair_comparison]	GHG Stmt Sec C
2.5	Injectate mass uses the conservative value (rail: scale tickets; loaded mass never over-drawn)	Flag	GPRX5102 (4 offloads 77,818.24352 kg against 77,940 kg loaded, loss 121.75648 kg) and TILX200231 (77,455.36992 kg against 77,520 kg, loss 64.63008 kg) are fully drawn below their BOL totals, so the last-offload back-calculation is not triggered and the raw ticket nets stand (Failure Category 11; mass-chain last_offload_derivation PASS on both). SHPX215046's four offloads, 1-2 here (21,055.74064 + 20,992.23776 kg) and 3-4 in W2 (20,810.80096 + 14,841.53024 kg, W2 live Ops Notes F28:F29), total 77,700.3096 kg against 77,680 kg loaded (BOLs 2462-R/2463-R/2464-R), 20.3096 kg more than the railcar held. The back-calculation applies to the last load, offload 4 (full 2026-09-07 10:55, empty 2026-09-08 07:07), which W2 booked: its value is 14,821.22064 kg, the figure COBB row 1731 carries. This removal's offloads 1-2 are correctly at their raw ticket nets. W2's Certify entry, read today, still books the raw 14,841.53024 kg, so the railcar stays over-credited by 20.3096 kg (0.029641 tCO2e at W2's carbon 40.4 %) until W2's offload 4 is corrected; W2's checklist fails the same 20.3096 kg. W2's railcar note M12 also says its offload 3 ticket 'is confirmed mislabeled' and 'LD4 has no resolvable ticket'. [log:mass_chain.shpx215046_balance, mass_chain.last_offload_GPRX5102, mass_chain.last_offload_TILX200231, ops_notes.railcars, cobb.w2_sister_rows_SHPX215046, w2_live_ops, w2_certify_today, policy.last_load_ruling, recompute.findings]	GHG Stmt Sec C
2.6	Sparging deduction (rate 0.0129) applied correctly	Pass	Ops D46 2,545.448535 kg is 0.0129 of C46 197,321.5918 kg (constant O5 0.0129), applied per row in H28:H37; recompute section 2 reproduces it. [log:ops_notes.rollup.sparging_kg, ops_notes.rollup.mass_used_kg, ops_notes.constants, recompute.lines]	GHG Stmt Sec C
2.6a	Filtration loss deduction applied correctly (weekly methodology)	Pass	Ops E46 381.7612409 kg is 0.00196 x (C46 - D46). Constant O7 0.00196 hyperlinks to 'AECN COA Solids Content Dataset (May 2025 - May 2026)', opened this session: Summary B12 'Recommended value for filtration loss' 0.001959191176, the per-load mean over 136 loads and 71 certificates. [log:ops_notes.rollup.filtration_kg, ops_notes.constants, filtration_constant_source, recompute.lines]	Weekly methodology
2.7	All mass values consistent across scale tickets/BOLs, Ops Notes and Certify	Fail	Certify 'Mass of Injectant (per injection)' is 194,394.38206 kg. The Ops Notes it cites gives MOBI F46 193,491.9399 kg (C46 - D46 - E46 - K46), and the Ops Notes copy attached in Certify shows the same F46. The difference, 902.44216 kg, is K46 'Mass Removed for R&D' (R4 200 gal x R5 3.785411784 L/gal x R3 1.192 kg/L = 902.4421693 kg): Certify deducts sparging and filtration but not the R&D removal, so it credits 902.44 kg of oil that was not injected. The removal itself is recorded only as R4 '200' under 'Gallons removed': the Ops Notes gives no date, offload or purpose for it and the September guidance does not mention it, so once Certify deducts it a verifier will also need that record. 'Bio-Oil Mass - AECN via Rail' 197,321.59 kg correctly equals the pre-deduction C46. [log:certify.components, certify.sources_downloaded_and_opened, ops_notes.rollup.mobi_kg, ops_notes.rollup.mass_removed_rnd_kg, ops_notes.constants, ops_notes.rnd_record, _derived_values.moi_delta_calc]	VR Sec 5.1
2.8	*Direct trucks: BOL-as-mass methodology*	*N/A*	*No direct trucks this week: the AECN Truck tab rollup A28 is 0 and rows B10:B24 are blank. [log:ops_notes.aecn_truck_tab]*	
3. Bills of Lading & Transportation				
3.1	Truck BOL(s) present for each shipment	Pass	Nine inbound truck BOLs, one per AECN-to-SOPOR shipment: 2456-R, 2457-R, 2458-R (GPRX5102); 2462-R, 2463-R, 2464-R (SHPX215046); 2468-R, 2469-R, 2470-R (TILX200231). Each is a 5-page PDF (BOL, shipping details, BioEnergie weigh ticket, TSCA import certificate, Charm PO Q42025) downloaded from the Ops Notes links and read; all 9 are in Certify Sources, byte-identical, linked to both mass datapoints. [log:bols.per_bol, bols.per_railcar, certify.bol_sources_compare, certify.attachment_map]	VR App. 1
3.2	Rail: 3 BOLs per railcar, destination SOPOR railyard	Pass	Three BOLs per railcar, each shipping to 'CHARM Industrial Inc. /WATCO, Société du port ferroviaire de Baie-Comeau (SOPOR), 18 Route Maritime, Baie-Comeau QC'; ship dates 2026-08-03 to 2026-08-11. [log:bols.per_bol, bols.per_railcar]	VR App. 1

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

3.3	BOL mass matches or exceeds reported injectate mass	Flag	Railcar level, on the weigh-ticket 'Masse nette' basis the Ops Notes uses: GPRX5102 77,940 kg and TILX200231 77,520 kg exceed their offloaded totals. SHPX215046's 77,680 kg is exceeded by 20.3096 kg until W2's offload 4 is cut back (2.5); this removal's share of that railcar is correct. The September guidance also sets a different basis for this week, 'Loaded mass (kg) = BOL volume at 60°F (L) × 1.22 kg/L', giving 79,298.78 kg for GPRX5102 and 79,030.38 kg for TILX200231 (SHPX215046 keeps W2's metric-ton basis under the guidance's multi-week clause). Ops E10:E12 use weigh-ticket masses instead, so the Ops Notes and Certify do not follow the guidance a verifier is handed; on the guidance basis the two full railcars would show losses of 1,480.54 kg and 1,575.01 kg against the 121.75648 and 64.63008 kg shown. The weigh-ticket basis is the conservative one: 1.22 kg/L (FH 2026 LCA B6) is above every measured density on file (BOLs 1.1967 to 1.1991 g/mL at 60°F, SGS 1.192, vendor LCA 1.1997), and the Ops allocation it drives carries more embodied distance. Loaded mass enters Certify through the AECN-SOPOR allocation fractions and SHPX215046's rail allocations (Ops M32/M33, offload mass / 77,680 x 2,199). The guidance and the Ops Notes must be made to agree. [log:policy.last_load_ruling, bols.per_railcar, vvb_guidance.quote_method, vvb_guidance.quote_scope, vvb_guidance.quote_multi_week, ops_notes.railcars, ops_notes.offload_table, lca_files.fh_2026, lca_files.vendor_ci, lab_results.density_15C_g_cm3, mass_chain.shpx215046_balance, _derived_values.bol_density_range_calc, _derived_values.guidance_basis_loss_calc]	VR Sec 3.5
3.4	Origin and destination correct on all BOLs	Pass	All 9 BOLs ship from 'BioEnergie AE Côte-Nord Canada, 175 Portage des Mousses, Port-Cartier QC' to SOPOR Baie-Comeau, carrier Transport Baie-Comeau; the rail leg SOPOR to Opelousas is Watco (COBB Loads tab); the last leg runs from the Opelousas railyard to Basco (route map). [log:bols.per_bol, bols.cross_check, cobb.loads_tab, route_evidence.opel_basco_map]	PDD Sec A
3.5	Transportation distances consistent with known AECN routes and evidenced	Fail	Evidence linkage is the Fail: each of the three allocated-distance datapoints ('Allocated Distance AECN to Sopor Quebec (truck)', 'Allocated Distance Sopor Quebec to Opelousas (rail)', 'Allocated Distance Opelousas to B6 (truck)') carries only the Ops Notes, and the route maps and the Watco email are linked to the full-distance datapoints only; the allocated distances feed the three embodied components, so their mileage basis is not attached where it is used, and the attachment gate fails on all three. W2 had the reverse gap (maps linked, Ops Notes not). Values: AECN-SOPOR 107.49722 mi from the 173 km map (173 / 1.609344); rail 2,199 mi from the Watco email 'The total milage for this move was 2199.'; Opelousas-Basco 42.9 mi, the longest of three map routes. Allocated rail 5,588.31 mi matches Ops M38 5,588.31288 (2,199 for each FULL railcar plus 2,199 x SHPX215046's W3 share) and Opelousas-Basco 858 mi matches N38 (42.9 x 2 x 10). Three further gaps: (1) Certify's AECN-SOPOR allocated 818.69019 mi does not reproduce from Ops L38 818.7957175 mi (the Ops method with 107.49722 mi gives 818.77454), 0.10553 mi apart, effect 0.000025 t; (2) railcar notes M10 (GPRX5102) and M12 (TILX200231) state 'Mileage (AECN->SOPOR / Rail / Opel->Basco) intentionally left BLANK -- no independently sourced route-mileage reference found for this railcar', yet J/K/L on both rows hold 107.5 / 2199 / 42.9 and feed the allocations, so the attached Ops Notes tells a verifier its own mileage is unsourced; (3) the Watco email names no railcar, route or date, so it cannot tie 2,199 mi to these three moves. [log:certify.attachment_gate, certify.attachment_map, route_evidence.aecn_sopor_map, route_evidence.watco_email, route_evidence.opel_basco_map, ops_notes.rollup, ops_notes.railcars, certify.components, recompute.findings]	Transport Emissions Acctg v1.1
3.6	Transport mode matches oil type and route	Pass	Six transport components: AECN-to-SOPOR tanker truck (process + embodied), SOPOR-to-Opelousas rail (process + embodied), Opelousas-to-Basco tanker truck (process + embodied). [log:certify.components]	Certify Datapoints

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

3.7	Billing documents present and consistent with BOL	Flag	Each BOL PDF bundles the Straight BOL, the Shipping Details, the BioEnergie weigh ticket, the TSCA certificate and a Charm PO; all 9 read, and each weigh-ticket net equals the Ops Notes mass. Hygiene: the PO on page 5 of every bundle is Q42025 of 10/2/2025 (line 'Q32025', Ship To El Dorado, KS; see 1.5), and its seller signature is blank on 2462-R, 2463-R, 2464-R, 2468-R, 2469-R and 2470-R; 2457-R ('Correction poids vide et litres finaux') has its 'Tanker Tare wt(empty) (Kg)' hand-corrected 19,282 to 19,290 and litres 21,673 to 21,666 while the printed 25.99 MT was not recomputed (weigh-ticket net 25,980 kg), and Ops M10 describes that correction in 'lb'; 2458-R carries a handwritten 'correction poid vide/empty truck' (page 1 tare 19,130 against page 2 19,460). COBB Loads tab still records 2457-R as 25.99 MT and GPRX5102 as 77.95 MT. [log:bols.per_bol, bols.cross_check, bols.po_page, cobb.loads_tab, ops_notes.railcars]	VR App. 1
4. Bio-Oil Composition & Testing				
4.1	CHN analysis PDF present	Pass	SGS Certificate of Analysis SR26-05599.002 (SGS order 5494492), page 2 of the two-page report, in Drive Testing/ and in Certify Sources (byte-identical), linked to Carbon Content together with the Ops Notes; read this session. [log:lab_results, drive_folder.testing, certify.attachment_map, certify.sources_downloaded_and_opened]	VR App. 1
4.2	Carbon content within expected AECN range	Pass	Carbon 39.7 % (m/m), ASTM D5291 Method C, marked '#' ('Result is outside of test method limits and/or analytical range used in method precision study'). MRV_AECN_Analysis.xlsx Summary Statistics (opened this session): mean 39.420350877193 %, SD 6.09332409018271 %, n 285, so the two-SD band is 27.23 % to 51.61 %; the batch sits 0.046 SD above the mean. Ops Summary E17 0.397 and Certify Carbon Content 39.7 % match. [log:lab_results.carbon_pct, lab_results.hash_footnote, carbon_range, ops_notes.summary_aecn, certify.components, _derived_values.carbon_z_calc]	Biomass Feedstock Acctg v1.2
4.3	Hydrogen and nitrogen values reported and reasonable	Pass	Hydrogen 7.0 % (marked '#'; dataset mean 6.87438596491228 %, SD 0.562848694875852 %); Nitrogen '<0.75' % (m/m), below the reporting limit. [log:lab_results.hydrogen_pct, lab_results.nitrogen_pct, carbon_range]	Biomass Feedstock Acctg v1.2
4.4	Lab report references correct batch/sample ID	Flag	CLIENT ID '9-2026-w3-AECN', SAMPLED 18-Sep-2026 17:00 (Ops Summary D17 composited 2026-09-18), RECEIVED 22-Sep-2026, report 23-Sep-2026: the identity ties. The LOCATION field prints 'FORT LUPTON, CO' on the AECN page, while the AECN composite is drawn at Basco, Louisiana; W2's certificate printed the same location. [log:lab_results.client_id, lab_results.sampled, lab_results.received, lab_results.location_on_certificate, lab_results.prior_week_location, ops_notes.summary_aecn]	VR Sec 5.1
4.5	*pH/conductivity (aqueous phase)*	*N/A*	*AECN is not aqueous phase. [log:oil_type]*	
4.6	Density above formation fluid threshold for safe injection	Pass	Density at 15°C 1.192 g/cm³ (ASTM D4052). Formation fluid: Waypoint Analytical report 24-200-0020, Lab No 98021, page 5, 'Specific Gravity 1.01 g/mL', ASTM D-1298-85 (opened this session). The bio-oil is denser than the formation fluid. [log:lab_results.density_15C_g_cm3, lab_results.formation_fluid]	UIC Permit / safe injection
5. Ops Notes Accuracy & Internal Consistency				
5.1	Correct batch ID and injection week in Ops Notes	Flag	'AECN Rail'!B2 reads INJ-09-2026-W3, week 2026-09-11 to 2026-09-18; all 10 offload rows dated 2026-09-18; Certify period 2026-09-12 to 2026-09-18, inside the guidance's W3 window '11 – 18 September 2026'. Label error: A3/A4 read 'Week Start (Sat)' and 'Week End (Sat)' but 2026-09-11 and 2026-09-18 are Fridays; W2's Ops week ran Saturday 2026-09-05 to Saturday 2026-09-12, so 2026-09-11 and 2026-09-12 fall in both weeks' windows (no offload is booked twice). [log:ops_notes.batch_id_cell, ops_notes.week_start, ops_notes.week_end, ops_notes.week_labels, ops_notes.offload_table, certify.removal_period, vvb_guidance.quote_scope]	verify header
5.2	Scale ticket values correctly transcribed into Ops Notes	Flag	Ops D/E equal the 20 ticket gross weights exactly. Ticket labels are inconsistent: GPRX5102 offload 2's full ticket 1124426253554 carries commodity 'SHPX215046 O4' and TILX200231 offload 4's full ticket 1124426255569 carries 'SHPX215046 3', both naming another railcar; W2 booked SHPX215046 offloads 3-4 on different tickets (1124426250545 and 1124426250546). TILX200231 labels also vary ('TIMX200231 O1', 'TILX2000231 O1', 'TILX2000231 O2'). [log:scale_tickets.pair_comparison, scale_tickets.w2_shpx215046_tickets, mass_chain.counts]	VR Sec 5.1
5.3	*Oil and aqueous phase mass reconciliation*	*N/A*	*AECN-only removal, no aqueous phase. [log:oil_type]*	

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

5.4	COBB values present and match Ops Notes/Certify	Flag	COBB tab 'Basco Injection – COBB' rows 1776-1785 (import 2026-09-29 11:54:56), column A 'INJ-09-2026-W3_bio_oil' confirmed before reading: all 10 offloaded_quantity_kg equal the Ops nets (mass-chain cobb_offload_consistency 10 of 10 PASS), consumed_lot follows Rail_<railcar>_<date>_offload_N and BOL_number lists each railcar's three BOLs. Two issues: an earlier import of the same 10 rows sits at rows 848-857, where row 856 has quantity_consumed_in_injection 61,207.70448 kg against offloaded 20,402.56816 kg; origin_quantity_kg for GPRX5102 is 77,950 kg against the corrected BOL total 77,940 kg (Ops M10: 'known break, not reconciled'). Neither reaches Certify. [log:cobb.tab, cobb.row_id_at_qa, cobb.batch_id_col_a, cobb.offloaded_quantity_kg, cobb.origin_quantity_kg, cobb.duplicate_block_difference, ops_notes.railcars, mass_chain.counts]	COBB tracker
5.5	pH measurement noted	Fail	pH is blank on all 10 offload rows (G28:G37), in the rollup G46, and in COBB injection_ph on all 10 rows. pH is required for every batch; W2 had the same gap. pH is not a Certify datapoint, so Net CDR is unaffected. [log:ops_notes.ph_column_G28_G37, ops_notes.rollup.pH_avg, cobb.injection_ph_all_rows]	GHG Stmt Sec C
5.6	Completion date present and matches COBB	Pass	COBB injection_completion_date 2026-09-18 on all 10 rows; Ops offload rows dated 2026-09-18; Certify period ends 2026-09-18. [log:cobb.injection_completion_date, ops_notes.offload_table, certify.removal_period]	VR Sec 3.5
5.7	*No erroneous density measurements*	*N/A*	*Density field expected blank for AECN in Ops Notes; the lab density is used only for the safe-injection check (4.6). [log:oil_type]*	
6. AECN Feedstock Emissions & Allocation				
6.1	AECN process emissions LCA CI spreadsheet present	Pass	'Bioenergy AE Cote-Nord Canada - LCA CI calculation - 1Jan2026 to 30Jun2026.xlsx' is in Certify Sources, linked to the pyrolysis-process CI datapoint; downloaded and opened: Summary!E19 0.13305202780074143 t CO2eq/t, period 2026-01-01 to 2026-06-30. [log:lca_files.vendor_ci, certify.attachment_map]	VR App. 1
6.2	LCA with stack emissions file present	Pass	'FH 2026 LCA w Stack Emissions.xlsx' is linked to the same datapoint; Sheet1!I12 'Total Process Emissions MT CO2e/MT Bio Oil' 0.1371171258 (stack I10 0.004065098048 + vendor I11 0.133052027800741). [log:lca_files.fh_2026, certify.attachment_map]	VR App. 1
6.3	Supporting GHG data file present	Pass	'Supporting GHG data 30Jun2026 S.xlsx' is linked to the same datapoint; opened: 'Inputs for LCA CI calculation for the period 1 Jan to 30 June 2026', production 10,334,884.34 L. [log:lca_files.supporting_ghg, certify.attachment_map]	VR App. 1
6.4	AECN pyrolysis process EF matches current half-year value	Pass	Certify CI 0.13712 tCO2e/t against Standard EF sheet 2026 tab row 4 'Production Emissions for AECN Oil–H2 2026', Final Quantity 0.1371171258 (source 'FH 2026 LCA w Stack Emissions'); the removal falls in H2 2026. Certify's value is the sheet value rounded up at the fifth decimal, 0.000567 t more emission (recompute section 4). [log:standard_efs.row4_aecn_process_H2_2026, standard_efs.vs_certify.aecn_process, standard_efs.period_basis, recompute.findings]	Certify Component Inputs
6.5	AECN pyrolysis embodied EF applied	Pass	Certify CI 0.02116 tCO2e/t; Standard EF sheet row 5 'Embodied Emissions for AECN Oil' Final Quantity 0.02116; 'AECN Embodied Emissions Calc Sheet (1).pdf' (linked, opened) gives 0.02115941123. [log:standard_efs.row5_aecn_embodied, lca_files.aecn_ee_calc, certify.attachment_map]	Certify Datapoints
6.6	Mass basis for pyrolysis EFs is pre-sparging origin mass	Fail	Both pyrolysis components use 'Bio-Oil Mass - AECN via Rail' 197,321.59 kg, the sum of the 10 pre-sparging ticket nets (Ops C46), which covers the delivered oil. GPRX5102 and TILX200231 are fully drawn and lost 186.38656 kg between loading and offload; that oil was produced and was in Charm custody, so its pyrolysis emissions (0.029501 t) also belong to this removal and are not charged (8.2). [log:certify.components, ops_notes.rollup.mass_used_kg, ops_notes.railcars, policy.custody_rule, recompute.findings]	Certify Components
7. Gross-to-Net Calculation				
7.1	Gross CO2e = injectate mass x carbon wt% x 3.667	Fail	Certify applies the formula to its own inputs correctly (194.39438206 t x 0.397 x 3.667 gives 282.999147 t), but its mass input is 902.44 kg too high (2.7). From the Ops MOBI, 193.4919399 t x 0.397 x 3.667 gives 281.685373 t, so the sequestration is overstated by 1.313774 tCO2e. [log:certify.components, _derived_values.gross_internal_calc, recompute.findings]	GHG Stmt Sec E

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

7.2	Process emissions correctly deducted	Fail	Activities 46.439 t in the Calculation view equal pyrolysis 31.232061 t (27.056736 + 4.175325) plus bio-oil transport 15.206929 t (process 14.634650 + embodied 0.572279), and every component recomputes (9.2, 9.4). The total leaves out 0.042411 t of upstream emissions on the 186.38656 kg lost from GPRX5102 and TILX200231 (8.2); with it charged, activities are 46.481401 t. [log:certify.calc_view, certify.components, _derived_values.activities_calc, recompute.findings]	GHG Stmt Sec E
7.3	Uncertainty discount correctly applied	Pass	Calculation view: 'Net 249.637 = Sequestrations 282.999 + Adjustments 14.635 - Activities 46.439 - Uncertainty Discount 1.558'. Re-derived from the component inputs: sequestration one-sigma (mass +/- 725.7 kg on 194,394.38206 kg, carbon +/- 0.15883 on 39.7 %) combined with the mass-driven activities' one-sigma (+/- 725.7 kg on 197,321.59 kg) reproduces 1.5577199676 t (recompute section 8), 0.55 % of gross. [log:certify.calc_view, certify.uncertainty_discount_t, recompute.lines, _derived_values.discount_pct_calc]	Isometric Standard v2.0 S3.7
7.4	Net removals consistent with BCU Quant and GHG Statement	Flag	Certify's GHG statements list (read today) ends at '01 Aug 2026 to 31 Aug 2026', Certificates issued 10 Sep 2026; the September 2026 statement has not been created, so there is no RP-level BCU Quant or GHG statement to cross-check yet. [log:rp_status]	RP BCU Quant.xlsx
7a. Per-Component Uncertainty Verification				
7a.1	Layer 1 - every component that needs uncertainty has it on the right input	Pass	All 10 component detail views opened. Injection sequestration shows +/- on carbon and mass; both pyrolysis components and the three transport-process legs show +/- on their mass input; the three transport-embodied components are per-mile with no mass input; the BCU is a constant reduction. None carries +/- on a distance or factor input. [log:certify.modal_reads, certify.components]	MRV Uncertainty Cheat Sheet
7a.2	Layer 2 - reported +/- matches the expected value	Pass	Mass +/- 725.7 kg on every mass input: 10 offload weighings x 72.57 kg, one outbound weighing per offload (Failure Category 9). Carbon +/- 0.15883 % against 0.004 x 39.7 = 0.1588 %. [log:certify.modal_reads, certify.components, _derived_values.unc_calc]	Isometric Standard S3.7.3
7a.3	Layer 3 - discount present and consistent	Pass	Uncertainty Discount 1.5577199676 t is present in the Calculation view, carried into the net, and reproduces from the per-component inputs (7.3; recompute section 8). The cross-check against the GHG statement waits for the September statement (7.4). [log:certify.uncertainty_discount_t, certify.calc_view, recompute.lines, rp_status]	Isometric Standard S3.7.3
8. Data Integrity & Anomaly Check				
8.1	No unexplained outliers in mass, carbon or calculated values	Flag	Nets run 39,740 to 46,420 lb, full gross 68,080 to 75,300 lb, and carbon sits 0.046 SD from the mean. Three tickets split their axles unlike the rest: full tickets GPRX5102 offload 2 (1124426253554: steer 0, drive 25,680, trailer 45,580, gross 71,260 lb) and TILX200231 offload 2 (1124426254565: steer 0, drive 22,160, trailer 50,480, gross 72,640 lb), where every other full ticket reads 9,820 to 13,060 lb on the steer; and empty ticket TILX200231 offload 1 (1312226254521: steer 3,880, drive 16,900 lb), where the other empties read 9,420 to 10,640 lb on the steer. Each gross is the sum of its readings and sits in range, and every ticket states 'The GROSS WEIGHT is the CERTIFIED WEIGHT and was weighed on a full length platform scale', so the nets stand. [log:scale_tickets.offloads.GPRX5102_offload_2, scale_tickets.offloads.TILX200231_offload_2, scale_tickets.axle_notes, _derived_values.ticket_ranges_calc, _derived_values.carbon_z_calc]	GHG Stmt Sec C
8.2	Spills, losses and irregularities documented	Fail	GPRX5102 offloaded 121.75648 kg (0.156 %) less than loaded and TILX200231 64.63008 kg (0.083 %), recorded in Ops G10 and G12; the September guidance discloses no spill. Upstream emissions of all oil in Charm custody are netted against removals, but pyrolysis, AECN-SOPOR truck and rail emissions are charged on the offloaded 197,321.59 kg, so the lost 186.38656 kg carries none of them: pyrolysis 0.029501 t + truck 0.002290 t + rail 0.010620 t, 0.042411 t uncharged (Certify factors). The August guidance applied the same principle to the GPRX5139 spill: 'The full 77,800 kg carries its upstream burden — pyrolysis, truck to SOPOR, rail, and truck to Basco — while only the 74,380.92 kg that reached the wellhead earns a removal.' [log:ops_notes.railcars, vvb_guidance.august_spill_quote, vvb_guidance.no_other_sections, policy.custody_rule, recompute.findings, _derived_values.loss_upstream_calc]	GHG Stmt Sec C

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

8.3	Inventory reconciliation / carryover documented	Fail	SHPX215046 is drawn across two weeks and out of order: offloads 1-2 weighed 2026-09-06/07 are booked here, offloads 3-4 weighed 2026-09-07/08 were booked in W2. The guidance reads 'Where a railcar is offloaded across more than one reporting week, the loaded mass is recorded once, in the week the railcar's first offload occurs, and subsequent weeks draw against the same shipment.' W3 records the 77,680 kg loaded mass again with H11 'Previously Claimed' 0 against the 35,652.3312 kg W2 booked, so the attached Ops Notes shows the railcar PARTIAL (F11 42,047.9784 kg, G11 blank) although all four offloads are booked; with H11 at 35,652.3312 kg, F11 reads 77,700.3096 kg and the 20.3096 kg over-draw (2.5) shows. The attached record misstates the railcar balance a verifier would check; H11 should carry W2's draw, 35,652.3312 kg as booked today and 35,632.0216 kg once W2's offload 4 is cut back. [log:policy.last_load_ruling, ops_notes.railcars, mass_chain.shpx215046_balance, w2_live_ops, vvb_guidance.quote_multi_week]	VR Sec 3.5
8.4	Rail uncertainty conservatism: offloads at least equal inbound BOLs	Pass	GPRX5102 and TILX200231: 4 offloads each against 3 inbound BOLs; SHPX215046: 4 offloads (2 in W2, 2 here) against 3 BOLs. Per-offload uncertainty (7a.2) books more than the inbound weighings would. [log:mass_chain.rail_uncertainty_precondition, bols.per_railcar]	Isometric Standard S3.7.3
9. Isometric Certify - Removal Component Mapping				
9.1	SEQUESTRATIONS: gross injection sequestration matches internal gross	Fail	Injection sequestration 282.999147 tCO2e reproduces from Certify's inputs (7.1) but not from the evidence: the Ops MOBI gives 281.685373 tCO2e, 1.313774 t lower, because Certify's Mass of Injectant omits the 902.44 kg R&D removal (2.7). [log:certify.components, _derived_values.gross_internal_calc, recompute.findings]	Certify Components
9.2	ACTIVITIES - Transport: process + embodied correct for all legs	Fail	All six legs recomputed from the Standard EF sheet 2026 tab (tanker 0.0001143 MT/(t·mi) and 0.0002375605888 MT/mi; rail 2.5910374e-05 MT/(t·mi) and 3.113066508e-05 MT/mi): process 2.424477, 11.242775 (Certify 11.242613) and 0.967560 t; embodied 0.194514 (Certify 0.194488), 0.173968 (Certify 0.173964) and 0.203827 t. Certify carries the rail fuel factor as 0.02591 kg/(t·mi), the sheet's 0.025910374 rounded, so its rail process is 0.000162 t lower; every difference is under 0.001 t. Transport process 14.634650 t plus embodied 0.572279 t gives the Bio-oil transport subtotal 15.206929 t. The railcar embodied value is the 2026 figure (65.37 MTCO2e / 2,100,000 mi). The upstream process legs are charged on the offloaded 197,321.59 kg, so the 186.38656 kg lost in GPRX5102 and TILX200231 carries no AECN-SOPOR truck (0.002290 t) or rail (0.010620 t) emissions (8.2). [log:policy.custody_rule, standard_efs, recompute.findings, recompute.lines, route_evidence.railcar_ee_pdf, _derived_values.transport_calc, _derived_values.activities_calc]	Certify Components
9.3	*ACTIVITIES - Injection: LCS / sample transport*	*N/A*	*AECN needs no LCS pre-treatment; none of the 10 components is an injection-side activity. [log:certify.components]*	*Certify Components*
9.4	ACTIVITIES - Pyrolysis: process + embodied entries present	Pass	'AECN pyrolysis (from rail loads) - process emissions' 27.056736 t (197.32159 t x 0.13712) and '... - embodied emissions' 4.175325 t (197.32159 t x 0.02116), both recomputed. [log:certify.components, recompute.lines]	Certify Components
9.5	REDUCTIONS: applicable BCU categories populated	Pass	'BCU - Transport Process', Constant CO_2 reduction 14,635 kgCO2e, sourced to Charm-RetirementCertificate-RD-EACs-20251218.pdf (opened: 2,795 Renewable Diesel TERCs retired Dec 18 2025, vintage 2024; 'Each TERC represents one metric ton of carbon dioxide equivalent ("tCO2e") reduced.'). Transportation Emissions Accounting v1.1 §4.0: 'the use of Book and Claim Units (BCUs) for low-carbon fuels to substitute for some, or all, of project fuel usage is permitted'. Only the transport-process category applies to this removal. [log:bcu_sources]	Certify Components
9.6	REDUCTIONS - BCU does not exceed transport process emissions	Fail	BCU 14.635 t against a transport-process sum of 14.634650 t (AECN-SOPOR 2.424477 t, rail 11.242613 t and Opelousas-Basco 0.967560 t, Certify values; 14.634812 t with sheet factors): the BCU exceeds the emission it offsets by 0.35 kg (0.19 kg at sheet precision). §4.2.1 Eq 5 allows at most the BCUs for which '(F_j − F_BCU,j ED_BCU,j/ED_Fuel,Transportation,j) ≥ 0'. The fix is the exact process figure, not a rounded one: 14,634.650 kgCO2e on today's activities, 14,647.560 kgCO2e once the lost oil's truck and rail emissions are charged (8.2), 14,647.392 kgCO2e with the 2.3 standard empty weight as well. [log:bcu_sources, policy.bcu_fix_rule, recompute.findings, _derived_values.bcu_calc]	Certify Components / Calculation View
9.7	COUNTERFACTUALS: zero per protocol	Pass	The Calculation view carries no counterfactual term: Sequestrations, Adjustments, Activities and Uncertainty Discount only (7.3). [log:certify.calc_view]	Certify Components

Batch ID: INJ-09-2026-W3-AECN | Reporting Period: September 2026 RP, Certify period 12 Sep - 18 Sep 2026 (Ops week 2026-09-11 to 2026-09-18) | Reviewer: Claude QA Agent | QA date: 2026-09-29 | Injection Site: Basco 6 | Oil Type: AECN rail (GPRX5102 offloads 1-4, SHPX215046 offloads 1-2, TILX200231 offloads 1-4; no direct trucks) | Certify: rmv_1M3MMSA7X1S0R377 | Scope: AECN removal components only (the Kerry removal in the same Drive folder is separate)

9.8	Net removal matches internal calculation	Fail	249.637 t reproduces from Certify's own terms (recompute 249.637427). It does not reproduce from the evidence: with the R&D removal deducted (2.7), gross 281.685373 t, BCU at the process sum 14.634650 t, activities 46.438990 t and the discount re-derived on the corrected mass 1.553904 t give 248.327129 t, 1.309871 t below Certify. Charging the lost oil's upstream emissions (8.2) gives 248.297628 t, and the 2.3 standard empty weight as well gives 248.294746 t (recompute CORRECTIONS). [log:certify.calc_view, recompute.findings, recompute.lines, _derived_values.net_internal_calc]	Certify Calculation View
9.9	All datapoints have descriptive source names	Flag	'image (44).png' is a generic filename on the rail distance datapoint; the image is a Watco email screenshot naming no railcar, route or date. 'Standard Emission Factors + Calculations (8).xlsx' is uploaded twice (two source records, byte-identical). The Standard EF sheet cites 'GLEC V3.2 2025' for the tanker and rail factors while the attached framework is the 'v3.1 edition, revised and updated (2024)'; its rail value 16.1 g CO2e/t-km converts to the sheet's 2.5910374e-05 MT/(t·mi), so only the version label differs. [log:route_evidence.watco_email, certify.certify_sources_enumeration_note, standard_efs.glec_version_cited, route_evidence.glec]	Certify Datapoints
9.10	All Sources match files in the batch folder / Ops Notes links	Pass	43 source records (42 filenames) enumerated with their linked datapoints. Every file that also exists in Drive was downloaded from Certify and matched byte-for-byte: 20 scale tickets, 9 BOLs and the SGS certificate; the Certify Ops Notes matches the live sheet on the AECN Rail and AECN Truck tabs (its Summary differs only by later Kerry rows). The route images, LCA files, EF extracts, BCU certificate and GLEC framework are Certify-only and were opened there. [log:certify.certify_sources_enumeration, certify.ticket_sources_compare, certify.bol_sources_compare, certify.sources_downloaded_and_opened]	Certify Sources
SIDEBAR - internal record (not a Certify source)				
S.1	BCU certificate capacity and usage ledger	Flag	Charm EAC Verification Inventory tracker, BCU Inventory row 16 (13-47206-50000, the certificate in 9.5): Total CO2 Emissions Reduction 2,765.361153 MT, used 414.9274093, remaining 2,350.433744, so the 14.635 t is covered. The certificate states 2,795 TERCs at one tCO2e each, 29.638847 t more than the tracker total. The usage tab's last row covers August 2026; September draws, this removal's 14.635 t included, are not logged yet. [log:bcu_sources.inventory_row16, bcu_sources.certificate_terc_count, bcu_sources.usage_rows_49_55, bcu_sources.september_2026_usage_rows, _derived_values.eac_calc]	EAC tracker